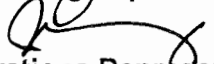


OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

STATION:	SALEM		
SYSTEM:	ADMINISTRATIVE		
TASK:	Review a completed surveillance to calculate Shutdown Margin		
TASK NUMBER:	1200030301		
JPM NUMBER:	15-01 NRC SRO A2		
ALTERNATE PATH:	☐	K/A NUMBER:	2.2.12
		IMPORTANCE FACTOR:	4.1
APPLICABILITY:		RO	SRO
EO ☐	RO ☐	STA ☐	SRO ☒
EVALUATION SETTING/METHOD:	Classroom		
REFERENCES:	SC.RE-ST.ZZ-0002, Rev. 23 Shutdown Margin Calculation S2.RE-RA.ZZ-0016, Rev. 8 Figures Salem Unit 2 Technical Specifications TSAS 3.1.3.1 Movable Control Assemblies		
TOOLS AND EQUIPMENT:	Calculator		
VALIDATED JPM COMPLETION TIME:	20 minutes		
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	30 minutes		
Developed By:	G Gauding Instructor	Date:	8-10-16
Validated:	R Boyer / P Williams SME or Instructor	Date:	9-28-16 / 10-4-16
Approved By:	 Training Department	Date:	10/11/16
Approved By:	 Operations Representative	Date:	10/11/16
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE:	☐ SAT	☐ UNSAT	
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:		DATE:	

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____

DATE: _____

SYSTEM: Administrative

TASK: Review a completed surveillance to calculate Shutdown Margin

TASK NUMBER: 1200030301

INITIAL CONDITIONS:

- Unit 2 Reactor Power is 100%.
- All Control Rod Group Demand Counters are at 227 steps.
- Current boron concentration is 300 ppm.
- Current core burnup is 10,000 EFPH.
- Control Rod 1D5 was declared INOPERABLE at **1000 today** due to not moving out during a rod exercise test. A Shutdown Margin Calculation has been performed IAW SC.RE-ST.ZZ-0002, Shutdown Margin Calculation, to satisfy the action requirement of TSAS 3.1.3.1 Action c.3.
- Control Rod 1D5 has been identified as **trippable** and is currently at 214 steps.
- The current time is 1030.

INITIATING CUE:

1. Review the attached SDM calculation procedure SC.RE-ST.ZZ-0002 for completeness and accuracy.
2. Note any discrepancies found by the review on Attachment 7.
3. Determine if SDM is SAT.

This is a time critical JPM. Time will start when procedures are given to you. Time will stop when procedures are returned to proctor.

Successful Completion Criteria:

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made

Task Standard for Successful Completion:

1. Determines the incorrect value for Control Bank Worth and Shutdown Bank Worth have been used.
2. Determines SDM with correct values used is SAT.
3. Completes review within 30 minutes.

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

System: ADMINISTRATIVE

Task: Review a completed surveillance to calculate Shutdown Margin

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
			Note to Evaluator: The time critical aspect refers to the time required when the Shutdown Margin requirement of TSAS 3.1.3.1.c must be met.		
	1	Provide the marked up copy of SC.RE-ST.ZZ-0002 and copy of S2.RE-RA.ZZ-0016, Curve Book. Time Start: _____	Ensures Sections 3 and 4 are completed.		
	2	Review Section 5 to ensure the correct Attachment was performed.	Reviews Section 5 and determines the correct Attachment (3) was performed.		
* *	3	Review the completed Shutdown Margin Calculation:	Reviews SDM and determines: 1. An incorrect value for Total Control Bank Worth (Step 4.2.1) has been used (0 EFPH value vs current 10,000 EFPH value) 2. An incorrect value for Total Shutdown Bank Worth (Step 4.2.2) has been used (0 EFPH value vs current 10,000 EFPH value)		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

System: ADMINISTRATIVE

Task: Review a completed surveillance to calculate Shutdown Margin

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
* * *	3 con't		3. Determines Trippable Rod Worth is incorrect based on incorrect steps 4.2.1 and 4.2.2) 4. Determines Ten Percent Rod Worth Penalty (step 4.3.2) is incorrect based on incorrect steps 4.2.1 and 4.2.2) 5. Determines SDM is incorrect based on incorrect steps 4.2.1 and 4.2.2 and 4.3.2) Correct Values are: 4.2.1 -3841 4.3.1 -5948 4.2.2 -3335 4.3.2 645 4.2.3 728 4.3.4 2742 4.2.4 0 4.3.5 -2441 4.2.5 500 4.2.6 0 4.2.7 -5948		
* *	4	Acceptance Criteria Time Stop: _____	Determines Acceptance Criteria are still met with correct values used. Turns in procedures within 30 minutes of start time. *****Total time to complete JPM MUST be 30 minutes or less.*****		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- ✓ 1A 1. Task description and number, JPM description and number are identified.
- ✓ 1A 2. Knowledge and Abilities (K/A) references are included.
- ✓ 1A 3. Performance location specified. (in-plant, control room, or simulator)
- ✓ 1A 4. Initial setup conditions are identified.
- ✓ 1A 5. Initiating and terminating Cues are properly identified.
- ✓ 1A 6. Task standards identified and verified by SME review.
- ✓ 1A 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- 1A 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 23, 8 Date 10/4/16
- 1A 9. Pilot test the JPM:
a. verify Cues both verbal and visual are free of conflict, and
b. ensure performance time is accurate.
- _____ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
- _____ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: [Signature]

Date: 9/28/16

SME/Instructor: [Signature]

Date: 10/4/16

SME/Instructor: _____

Date: _____

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

**INITIAL
CONDITIONS:**

Name: _____

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